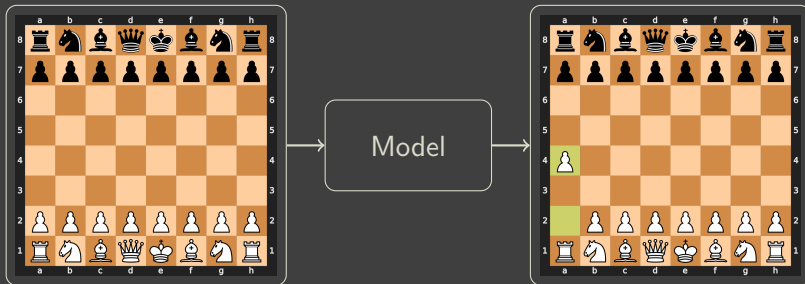


Exploration of Reward Shaping and Imitation Learning in Chess

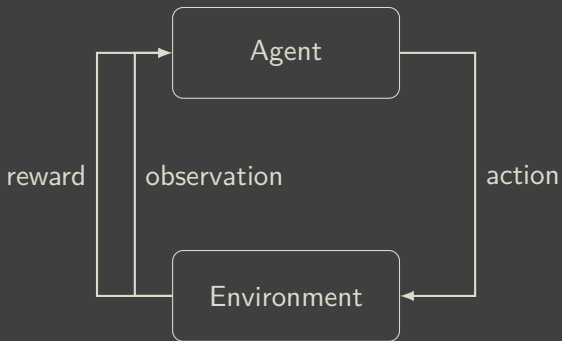
Jakob Lambert-Hartmann

2025

Introduction



Reinforcement Learning



REINFORCE

- **Goal:** Directly optimize a stochastic policy $\pi_{\theta}(a \mid s)$ by maximizing the expected return

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^T R_t \right]$$

- **Core idea:** Increase the probability of actions that lead to high returns

$$\nabla_{\theta} J(\theta) = \mathbb{E} \left[\sum_{t=0}^T \nabla_{\theta} \log \pi_{\theta}(a_t \mid s_t) \cdot G_t \right]$$

- **Algorithm:**
 - Sample trajectories using the current policy π_{θ}
 - Compute the return G_t for each time step
 - Update parameters via gradient ascent

Problems with rewards

- Sparse rewards → poor performance
- Reward Shaping → difficult:
 - Missaligned rewards
 - Reward hacking
 - Prioritizing wrong rewards

Imitation Learning: Behavior Cloning

- Supervised learning
- Agent learns expert demonstrations
- e.g. puzzles, grandmaster games

Problems:

- Demonstration sample bias → poor performance

Combination

1. Pretrain: Behavior Cloning
2. Finetune: Reward Based Reinforcement Learning

Can both approaches stabilize each other?

- Behavior Cloning: Stabilize RL with sparse rewards
- Reinforcement Learning: Compensates for data bias

Lets Find Out

- Train different models:
 - Architecture: Fully connected, cnn, residual block
 - Pretraining Data: Heavily biased $\rightarrow$ less biased
 - Rewards Sets: Sparse $\rightarrow$ Dense

Methods

Architecture

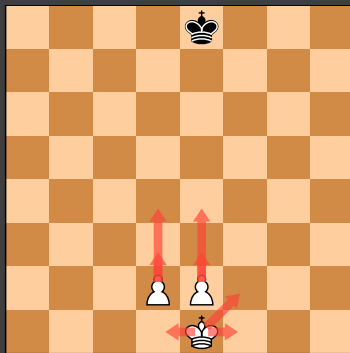
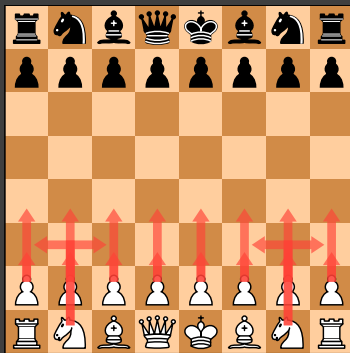
Input:

- tensor shape = (8,8,12):
 - 8 · 8 squares
 - 12 pieces:
 $\{Pawn, Rook, Knight, Bishop, Queen, King\} \times \{White, Black\}$

Output:

- Move distribution: from square $\rightarrow$ to square
- **But:** Not all moves are legal $\rightarrow$ mask illegal moves

Masking



- mask illegal moves
- masked logits $\rightarrow$ move distribution
- sample from move distribution

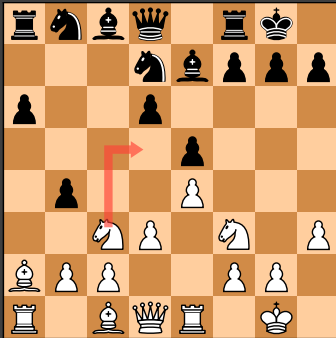
Architecture

- Fully Connected NN
 - 3 hidden layers: 512 neurons
- Convolutional NN
 - increasing number of channels:
 - $12 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 32$
- Residual Block NN
 - 20 blocks with 64 channels

Behavior Cloning

A good model needs to master strategy and tactics
(important to understand data bias):

Strategy:

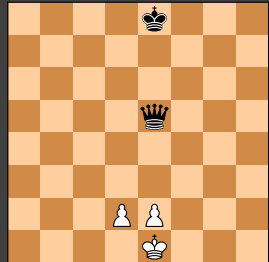
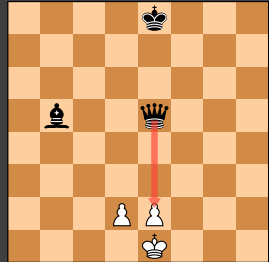


Tactics:



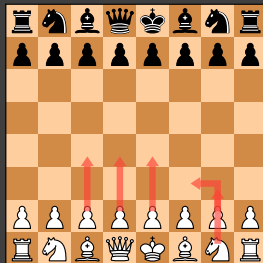
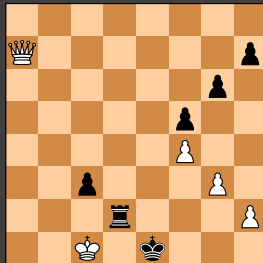
Dataset: Chess Puzzles

- lichess puzzle dataset
- 6 million positions
- **bias:**
 - no strategic positions
 - no positions without tactic



Dataset: GM Games

- chess.com master games
- balance tactics/strategy
- **bias:**
 - high level play
 - many resignations
- **other problem:**
 - multiple moves possible



Behavior Cloning

- Train model with one of both datasets
 - Save best after 1 epoch
 - Save best after 10 epoch

Reinforcement Learning: Reward Shaping

- Reward sets:
 1. Sparse: only reward on win/loss
 2. Medium: adding rewards material advantage, center control, king safety
 3. Dense: adding rewards outer center control, castling, pawn promotion, blunder prevention, checks

Selfplay: 1000 batches with 16 games each (REINFORCE)

→ total of 16000 games per color per model

Total Models

- Architecture:
 - fully connected
 - cnn
 - residual block
 - Pretraining:
 - untrained
 - puzzle data: 1 epoch
 - puzzle data: 10 epoch
 - gm data: 1 epoch
 - gm data: 10 epoch
 - Reward Sets:
 - none
 - sparse
 - medium
 - dense
- 57 models

Results

Behavior Cloning

dataset	epochs	fc	cnn	resblock
puzzle	1	62.0%	69.9%	74.6%
	10	65.8%	73.3%	81.7%
grand masters	1	31.0%	38.5%	41.5%
	10	31.8%	39.6%	43.7%

- Puzzle better generalizing
- **But** master games: different labels, same position

Evaluate

- For each architecture: play tournament with 1000 games per matchup

Reward Shaping

Best Model

architecture	train	rewards	points%
cnn	gm_10	r_0	67.4
fc	pz_1	r_0	17.6
resnet	pz_10	none	65.0

well... it's better than random (probably)